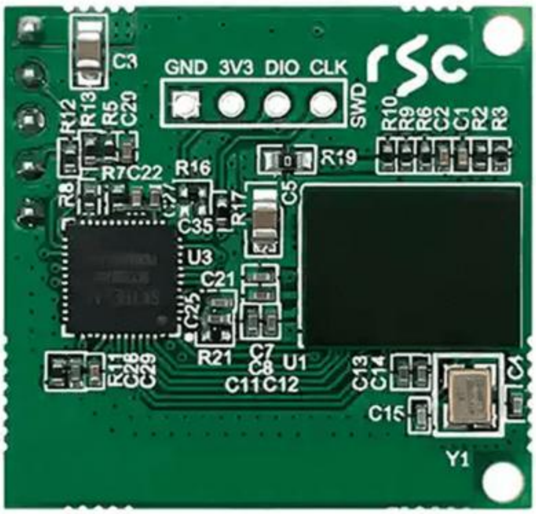
Shenzhen Hi-Link Electronic Co., Ltd.

HLK-LD6001A-60G Human Trajectory Radar Sensor Module Manual



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Contents

1. Product features

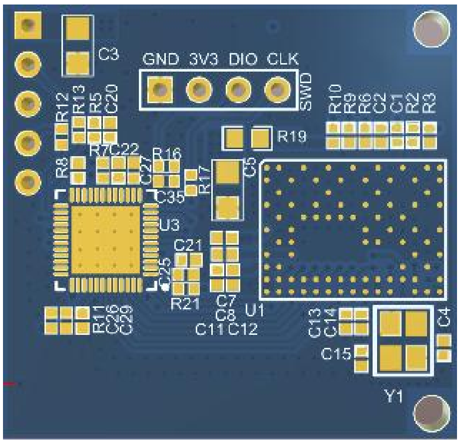
Compared to traditional sensing methods like vision, infrared, and laser, millimeter-wave radar is unaffected by lighting conditions. It enables round-the-clock passive monitoring of indoor personnel while protecting personal privacy, making it the optimal sensor for home environments. This product utilizes fully domestically developed chips for autonomous control, achieving precise tracking and positioning of multiple individuals indoors. It can detect stationary states such as reading or sleeping, while effectively suppressing interference from curtains and greenery. The system boasts advantages including low cost, complete domestic components, high reliability, and superior performance.

Items	Function	Detailed introduction
1	multiple target tracking	1) It can realize the target tracking function of up to 10 people, including the target motion trajectory and real-time position of the target; 2) Strong ability to suppress false targets (curtain, green plants, multipath, etc.); 3) High sensitivity to detect micro moving targets (standing still, shaking, waving, etc.).
2	zoning	Users can flexibly configure the detection area

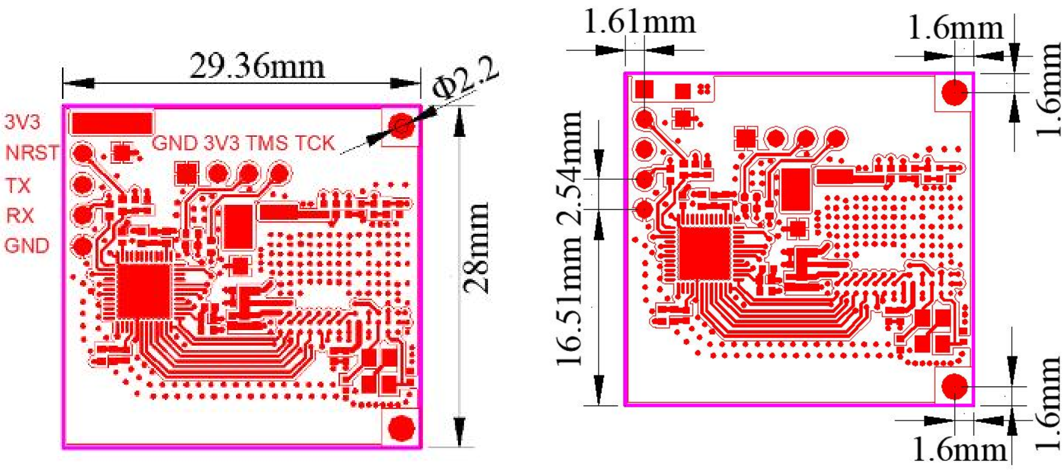
2. Product parameter introduction

Items	Parameter	Items	Parameter
1	Installation: top installation	6	Peak power consumption: 1.7w
2	Detection distance: 0.5~8m (The effective projection ground is a circle with a radius of 3.5m and the installation height is 2.7m)	7	Communication mode: TTL port
3	Azimuth and pitch Angle coverage: ±60°	8	Working frequency: 60-64GHz
4	Power supply: 3.3V	9	Processing cycle: ≤30ms
5	Average power consumption: 0.3w	10	Size: 29.36*28mm

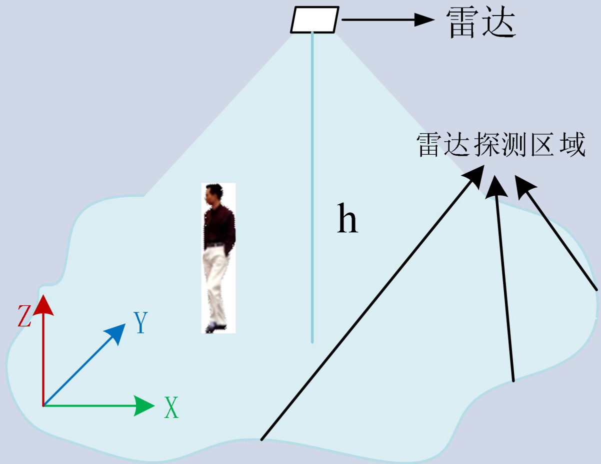
2.1. Product appearance



2.2. Product size



3. Product features introduction

Items	Characteristic	Detailed introduction
1	Installation scenarios	 Detection distance: 0.5~8m **Note: The detection distance is related to the installation environment, human body size, relative Angle and movement amplitude, etc. The above parameters are the test results of our company. Under different test conditions, the actual test results shall prevail.**
2	Not environmentally affected	Not affected by temperature, humidity, dust, light, noise and so on.
3	Flexible parameter configuration	The detection range and functional mode can be configured through the serial port.

4. Electrical character

4.1. Pin description

Pin	Description
3V3	Module power supply input
NRST	Module reset
TX	Serial port transmission
RX	Serial port reception
GND	The earth

4.2. Rated limit parameters

Pin	Min. value	Max. value	Unit
3V3	-0.5	3.6	V
I/O (TX/RX/NRST)	-0.5	3.6	V

4.3. Typical operating parameters

Pin	Typical value	Unit
3V3	3.0 ~ 3.3	V
I/O (TX/RX/NRST)	-0.5 ~ VDD+0.3	V

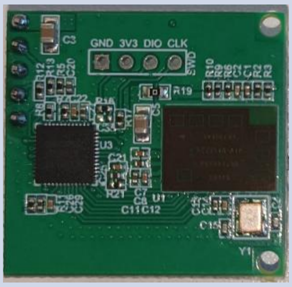
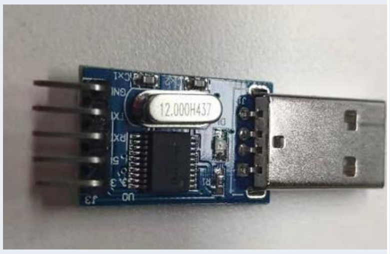
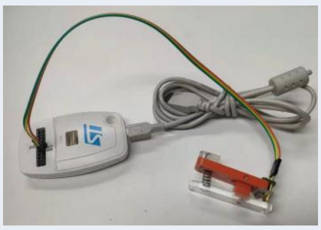
Note: VDD in the above table refers to the power supply input

4.4. Module dissipation

The radar module contains RF components. During active transmission periods, it consumes approximately 530mA of current, while during standby periods, the current drops to around 80mA. The average power consumption correlates with the radar's processing cycle duration: when operating at 100ms intervals, the average current reaches about 110mA. For power supply requirements, a high-capacity power supply is needed, with an output current of no less than 1A.

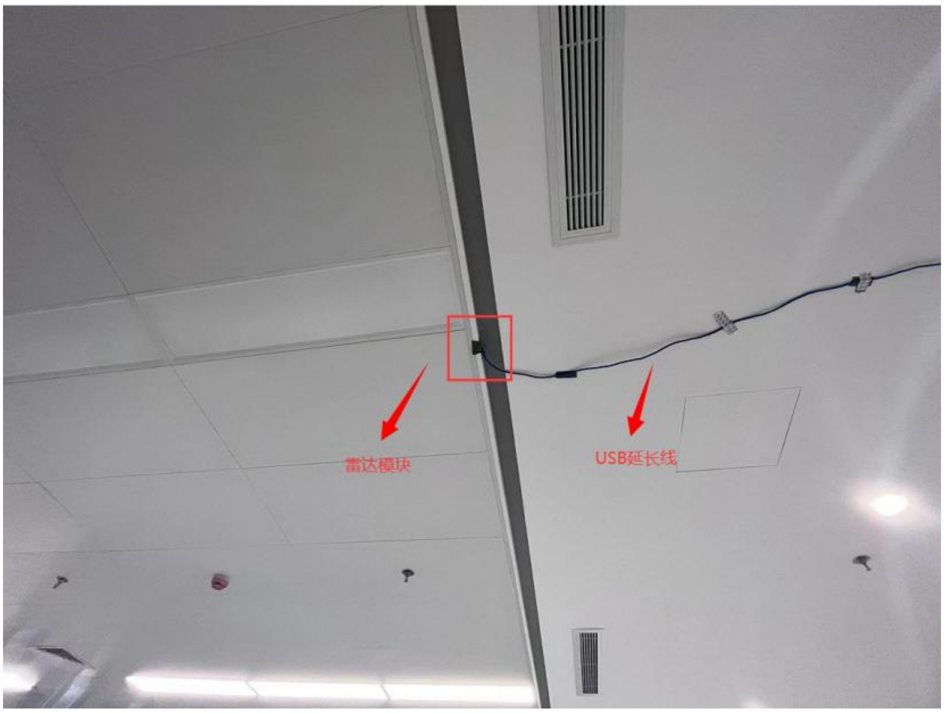
5. Environment building

5.1. Hardware composition

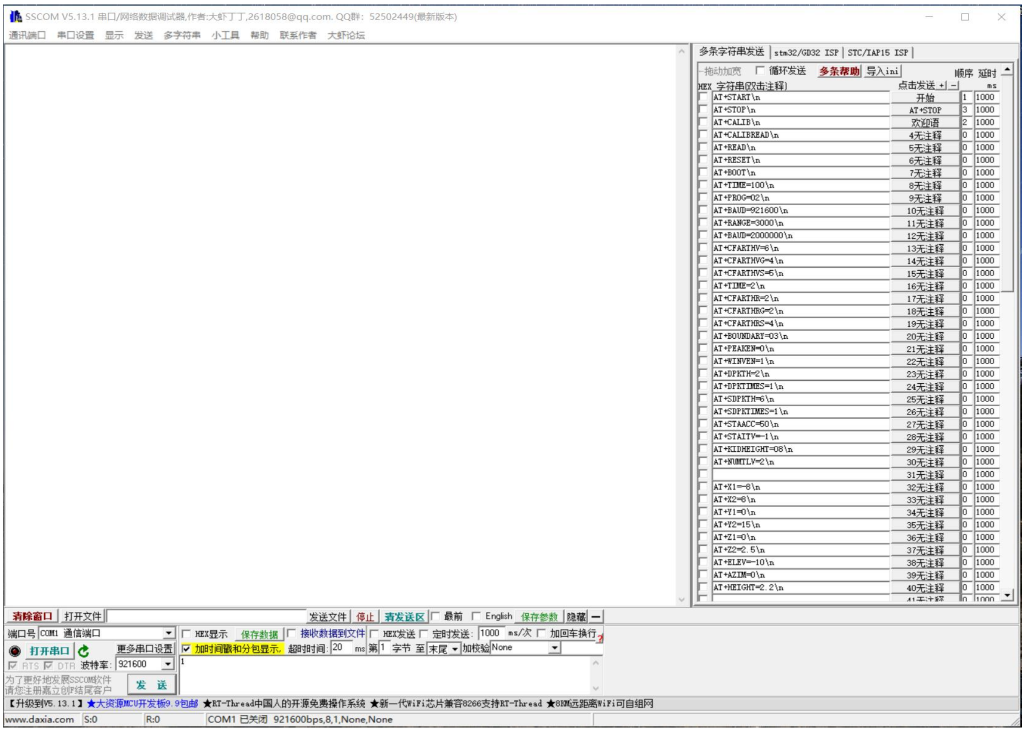
Items	Name	Picture	Description
1	Radar module		Model: HLK-LD6001A-60G
2	USB to TTL module		The USB to TTL module can realize serial command configuration, antenna calibration and other functions.
3	USB extension line		USB extension cable, a cable used to connect a PC to a USB-to-TTL module.
4	ST-LINK Downloader		ST-LINK downloader, used for radar module firmware upgrade, secondary development simulation and debugging. (optical)

5.2. Installation site

The module is installed on the ceiling antenna facing down, and the installation height is 2.5-3.0m. When installing the module, try to keep it fixed to avoid module shaking, and try to keep the surrounding environment open. The USB extension cable should be fixed as far as possible to avoid interference caused by the line.



6. Parameter configuration



Adjust the corresponding parameters according to the need. Note: After modifying the parameters, click the button behind the parameters to complete the modification.

Common parameter Settings are as follows:

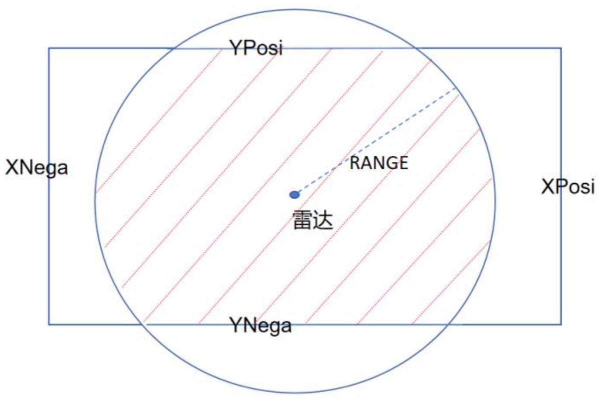
Commands	Meaning
AT+START\n	Start-up
AT+STOP\n	Stop
AT+RESET\n	Reset
AT+READ\n	Read parameters
AT+RESTORE\n	Restore to default settings
AT+DPKTH=X\n	Long-distance detection sensitivity (default is 4, range 1~9, the larger the value, the lower the sensitivity)
AT+BAUD=XX\n	Configure the serial port baud rate (default value is 921600)
AT+HEATIME=XX\n	Configure the heartbeat interval of the protocol output (unit: s, range: 10-999, default value: 60)
AT+RANGE=XX\n	Configure the radius of the ground radar detection circle projected by radar (unit: cm, range: 10-500, default value: 450)

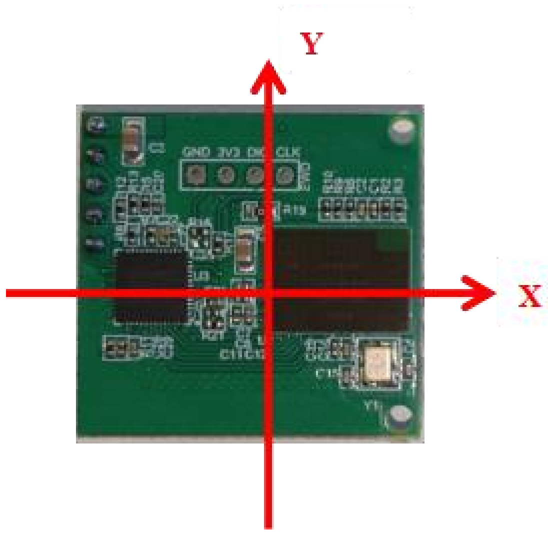
AT+HEIGHTD=XXX\n	Set vertical distance (unit: cm, setting range: 50~500, default value: 300)
AT+DEBUG=X\n	0: Protocol mode (default, simple protocol) 1: Output the string 2: Debug mode (used by the host computer) 3. Protocol mode (detailed protocol)
AT+XPosiD=XXX\n	Configure the positive range of X (unit: cm, range: 20 ~500, default value: 450)
AT+XNegaD=-XXX\n	Configure the X negative range (unit: cm, range: -500 ~ -20, default value: -450)
AT+YPosiD=XXX\n	Configure the positive range of Y (unit: cm, range: 20-500, default value: 450)
AT+YNegaD=-XXX\n	Configure the negative range of Y (unit: cm, range: -500 to-20, default value: -450)
AT+Moving=XXX\n	Configure the moving target disappearance time (unit 100ms, range 5~1000, default value is 110)
AT+Static=XXX\n	Configure the static target disappearance time (unit 100ms, range 5~1000, default value is 100)
AT+Exit=XXX\n	Configure target exit boundary time (unit 100ms, range 2~1000, default value is 5)

Note: If the configuration is successful, AT+OK will be returned. If the configuration fails, Save Para Fail will be returned, and the command needs to be sent again.

Border configuration description:

AT+RANGE=XXX\n, AT+XNegaD=XXX\n, AT+XPosiD=XXX\n, AT+YNegaD=XXX\n, AT+YPosiD=XXX\n are all boundary configuration instructions. Specifically, AT+RANGE specifies the radius of the ground radar detection circle projected by the radar, meaning the target's position must be within the distance defined by AT+RANGE. The area enclosed by these boundaries constitutes the radar detection zone. The schematic diagram is as follows:





7. Protocol specification

7.1. In protocol mode (AT+ DEBUG=0\n)

Example of a complete data packet:

55 AA 0A 04 00 00 00 00 00 0E

55 AA: Frame header

0A: Number of bytes

04: type=0x04 Number of people

00 00: Reserved

00 00: Reserved

00: Number of people counted

0E: XOR check (0A 04 00 00 00 00 00 XOR calculation)

7.2. In demo mode (AT+ DEBUG=3\n)

7.2.1. Total format for data upload

field		Number of bytes	explain
HEAD		8	The frame header is fixed as \x01 \x02 \x03 \x04 \x05 \x06 \x07 \x08
LENGTH		4	Length of the entire data (uint32)
FRAME		4	Frame number (uint32)
TLVs		4	TLVs=1（uint32）
POINTLENTH		4	The length of the point cloud is always 0 (uint32)
TLVs		4	TLVs=2（uint32）
TRACKLENTH		4	Personnel information length (uint32) (number of people = TRACKLENTH/32)
Personnel 1	F	4	Reserved
	ID	4	Personnel identification number (uint32)
	X	4	The X/Y/Z coordinates and speed of the personnel (X left/right, Y front/back, Z height) (float)
	Y	4	
	Z	4	

	Vx	4	
	Vy	4	
	Vz	4	
……		……	……
personnel n	F	4	Reserved
	ID	4	Personnel identification number (uint32)
	X	4	The X/Y/Z coordinates and speed of the personnel (X left/right, Y front/back, Z height) (float)
	Y	4	
	Z	4	
	Vx	4	
	Vy	4	
	Vz	4	
Check		1	Perform XOR verification of frame number and personnel information content

7.2.2. Example of a complete data packet

01 02 03 04 05 06 07 08 40 00 00 00 A3 01 00 00 01 00 00 00 00 00 00 00 02 00 00 00 20 00 00 00 00 00 00 00 00 00 00 00 21 28 96 BF CB 85 20 40 9A AB A3 3E 8A BD C1 3D 50 98 99 BD 40 52 C3 3A CC

01 02 03 04 05 06 07 08: Frame header

40 00 00 00: The total length of the frame is 64 bytes

A3 01 00 00: The current frame number is 419

01 00 00 00: TLV=1

00 00 00 00: Always zero

02 00 00 00: TLV=2

20 00 00 00: The length of the personnel data is 32, that is, the number of personnel =32/32=1 person

00 00 00 00: Reserved

00 00 00 00: The personnel ID is 0

21 28 96 BF 72 6F 81 BF CB 85 20 40: The XYZ coordinates of personnel 0 are-1.17,2.50, and 0.31 respectively

8 A BD C1 3D 50 98 99 BD 40 52 C3 3A: The XYZ velocity of personnel 0 is 0.094, -0.074, and 0.001

CC: A3 01 00 00 and personnel information 00 00 00 00 00 00 00 00 21 28 96 BF CB 85 20 40 9A AB A3 3E 8A BD C1 3D 50 98 99 BD 40 52 C3 3A are checked by XOR

7.3. Host usage (AT+ DEBUG=2\n)

Select the serial port in the figure below as COM74 and the baud rate is 115200. Click "Open serial port" and then click "Start" to run normally: